

TAMIL NADU STATE BOARD - STANDARD 9 MATHEMATICS

CHAPTER 9: PROBABILITY

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Mathematics
Chapter Name:	Probability

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

UNIT ROADMAP: PROBABILITY & EVENT SPACE



Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Random Experiment	An experiment where the outcomes are known, but the exact outcome of a specific trial cannot be predicted.
Sample Space	The set of all possible outcomes of a random experiment, denoted by S .
Event	A subset of the sample space S , representing a specific outcome or combination of outcomes.
Complementary Event	The event representing the non-occurrence of event A , denoted by A' , where $P(A) + P(A') = 1$.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

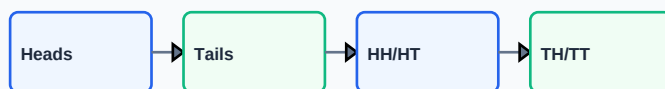
GLOSSARY: FUNDAMENTAL PROBABILITY TERMS



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: COIN TOSS TREE OUTCOMES PATH



Essential Formulas, Laws & Identities:

- Probability of Event E: $P(E) = n(E) / n(S)$
- Probability bounds: $0 \leq P(E) \leq 1$
- $P(\text{impossible event}) = 0$ | $P(\text{sure event}) = 1$
- Complement relation: $P(E') = 1 - P(E)$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A fair coin is tossed once. Find the probability of getting a Head.

Step-by-step Solution:

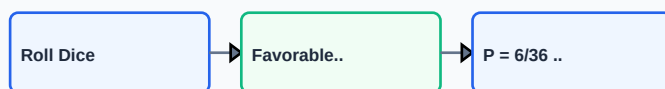
Sample space $S = \{H, T\} \Rightarrow n(S) = 2$. Event E (getting head) $= \{H\} \Rightarrow n(E) = 1$. $P(E) = n(E)/n(S) = 1/2 = 0.5$.

Example 2: A die is thrown. Find the probability of getting an even number.

Step-by-step Solution:

Sample space $S = \{1, 2, 3, 4, 5, 6\} \Rightarrow n(S) = 6$. Even numbers $E = \{2, 4, 6\} \Rightarrow n(E) = 3$. $P(E) = 3/6 = 1/2 = 0.5$.

WORKED: DICE PAIR PROBABILITY OUTCOMES



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Two fair coins are tossed simultaneously. Find the probability of getting at least one Head.

Analytical Solution Strategy:

Sample space $S = \{HH, HT, TH, TT\} \Rightarrow n(S) = 4$. At least one Head $E = \{HH, HT, TH\} \Rightarrow n(E) = 3$. $P(E) = n(E)/n(S) = 3/4 = 0.75$.

HOTS: NORMAL DISTRIBUTION PROBABILITY DENSITY



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Write the sample space when a single die is rolled.

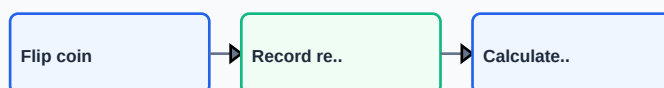
Q2 (2-Mark): Find the probability of drawing a red card from a standard deck of 52 cards.

Q3 (2-Mark): What is the probability of a sure event?

Q4 (5-Mark): If $P(A) = 0.65$, find $P(A')$.

Q5 (5-Mark): A bag contains 5 red and 3 blue balls. Find the probability of drawing a blue ball.

PRACTICE: EMPIRICAL PROBABILITY WORKSHEET



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Weather Forecasting

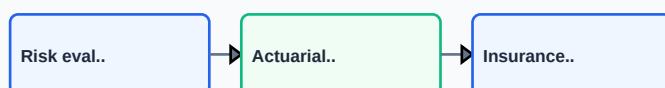
Weather systems probability models are used to calculate the probability of rain.

Application Case: Insurance Risk

Probability values are calculated to set premiums for risk management.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: PROBABILITY IN INSURANCE RISK ANALYSIS



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: PROBABILITY CONCEPTS SUMMARY

